

Wang Zijun

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Postdoctoral Fellow in Economics, HKU Business School | Ph.D., Renmin University of China

Education

University of Hong Kong (HKU)	Hong Kong SAR
Postdoctoral Fellow (Focus: Macro-Development & IO Micro-foundations)	2024 – Present
Renmin University of China (RUC)	Beijing, China
Ph.D. in World Economics (Focus: Open Macro, DSGE, Asset Bubbles)	2020 – 2024
M.A. in World Economics	2015 – 2017

Postdoctoral Research Experience

Postdoctoral Researcher (Industrial Organization & Macro-Finance) *HKU Business School*

- **Integrated Macro-Finance & Structural Modeling:** Developed an interdisciplinary framework bridging **DSGE** and **IO** models. Leveraged mathematical derivation to model heterogeneous agent optimization under uncertainty, explicitly linking firm-level strategic interactions (pricing/competition) to aggregate economic fluctuations.
- **Advanced Econometric & Computational Methods:** Mastered **Structural Estimation** techniques (e.g., BLP framework, Discrete Choice Models) using simulation-based methods (GMM, MLE). Focused on identifying causal mechanisms and demand elasticities in high-dimensional markets.
- **Alternative Data & Predictive Analytics (NLP):** Expanded methodological toolkit to include **Natural Language Processing**. Explored techniques to quantify firm-level sentiment and competitive pressure from unstructured text data to predict macroeconomic variables and asset pricing.

Job Market Paper

Asset Bubbles, International Capital Flow and Business Cycle *Ph.D. Dissertation*

- **Theoretical Innovation:** Constructed an asymmetric **Two-Country DSGE** model incorporating **Rational Asset Bubbles**. Modeled bubbles as "Sentiment Shocks" that trigger **Multiple Equilibria**, theoretically proving why capital flows "uphill" from high-growth EMs to DMs.
- **Mechanism Design:** Identified a transmission channel where asset bubbles induce over-investment, lowering the **Marginal Product of Capital (MPK)**. This challenges the neoclassical view that capital chases growth.
- **Quantitative Strategy (Bayesian):** Performed **Bayesian Estimation** using 40 years of quarterly data (CN/US). Conducted **Out-of-Sample Forecasting** and Variance Decomposition. Found that "Sentiment Shocks" explain 20%+ of capital flow variance.
- **Causal Identification:** Applied **Jordà's Local Projections (LP)** on panel data of 8 major economies to establish a robust causal link between housing price surges and FDI outflows.

Selected Working Paper

Housing Market Liquidity, Labor Misallocation, and TFP

- **Model Architecture:** Constructed a multi-sector DSGE-style endogenous growth model with three types of heterogeneous labor (Tech, Real Estate, Manufacturing) to quantify the non-neutrality of housing bubbles on long-term asset pricing and output.
- **Mechanism Design:** Modeled housing explicitly as both a production factor and a financial collateral asset. Derived the liquidity premium (M) analytically to explain how credit expansion distorts the relative price of labor, leading to a "hollowing out" of the manufacturing sector.
- **Dynamics & Calibration:** Solved a system of 18 dynamic structural equations to characterize the Balanced Growth Path (BGP). Calibrated model parameters using Chinese macro data to simulate impulse response functions (IRFs) under permanent financial shocks (LTV shocks).
- **Key Findings:** Demonstrated that while housing booms provide short-term liquidity, they induce a permanent labor structural change where real estate wages overshoot, crowding out R&D investment and lowering the steady-state TFP level, despite unchanged long-run growth rates.

Financial Asymmetry and Policy Spillovers: A DSGE Model of Strong Central Bank

- **Model Architecture:** Built an asymmetric Two-Country DSGE model (Core vs. Periphery) featuring a global banking sector to analyze the spillover effects of US monetary policy on Emerging Markets (EMs).
- **Key Frictions:** Incorporated Liability Dollarization and Financial Accelerator mechanisms. Modeled the "Strong Currency" effect where EM banks face a currency mismatch on their balance sheets, amplifying external shocks.
- **Dynamic Optimization:** Solved the banks' maximization problem under incentive-compatibility constraints to derive the endogenous leverage ratio. Analytically proved how US rate hikes tighten EM financial conditions via the net worth channel.

- **Policy Analysis:** Calibrated to China-US data (1992-2022) to evaluate policy counterfactuals. Found that Macro-prudential policies (e.g., LTV adjustments) in EMs can effectively mitigate 40% of the output volatility caused by US monetary tightening.

Macro-Prudential Policy and Global Monetary Policy Transformation

- **Theoretical Framework:** Developed a Two-Country New Keynesian model with a segmented financial market to analyze how "Hot Money" inflows crowd out the central bank's seigniorage rights under an inflation-targeting regime.
- **Novel Mechanism:** Identified an "Unregulated Third Market" (Real Estate) channel. Modeled how regulatory arbitrage drives capital into the housing sector when product markets are strictly regulated, explaining the divergence between low CPI inflation and high housing bubbles in China.
- **Policy Evaluation:** Evaluated the trade-off between capital controls and macro-prudential policies. Demonstrated that regulating the leverage ratio of the "Third Market" effectively restores monetary policy independence by reducing the elasticity of hot money to interest rate differentials.

Macro-Prudential Policy under Zero Lower Bound (ZLB)

- **Scenario Analysis:** Constructed a DSGE model with financial accelerators to evaluate policy effectiveness during deep recessions when the interest rate hits the Zero Lower Bound (ZLB).
- **Key Mechanism:** Analyzed the "Amplification Effect" of macro-prudential policy (LTV). Found that while LTV adjustments alone have limited impact during recessions, they significantly enhance the multiplier effect of Quantitative Easing (QE) by relaxing corporate collateral constraints.
- **Policy Implication:** Proposed a coordinated policy framework where counter-cyclical LTV adjustments should serve as a prerequisite "amplifier" for QE to be effective in stimulating real investment.

Publications

Age Structure and China's Economic Growth

Economic and Political Studies, 2024

Utilized IV-GMM to address endogeneity in panel data; analyzed demographic impact on long-term growth.

Population Aging and Household Risk Aversion

Applied Economics Letters, 2024

Quantified the non-linear impact of demographics on financial risk preferences using micro-survey data.

Work in Progress

Wealth Distribution and Property Tax: A Continuous-Time HANK Framework

- **Model Class:** Developing a continuous-time Heterogeneous Agent New Keynesian (HANK) model to quantify the wealth redistribution effects of property tax reforms.
- **Computational Method:** Solving the Mean Field Game (MFG) equilibrium system involving coupled Hamilton-Jacobi-Bellman (HJB) and Kolmogorov-Forward (KF) equations.
- **Algorithm:** Implementing a finite difference method with an implicit upwind scheme to handle high-dimensional state spaces and idiosyncratic income risk (diffusion processes).
- **Policy Goal:** Analyzing how housing acts as a wealth accumulation vehicle and evaluating whether property taxes can mitigate inequality without triggering a hard landing in the housing market.

Debt Service Burden and Variable Mortgage Rates: A Financial Intermediation Approach

- **Model Specifics:** Constructed a DSGE model with a Banking Sector subject to default risk (Gertler-Karadi type) and incentive compatibility constraints.
- **Core Mechanism:** Endogenized the Variable Mortgage Rate to analyze how interest rate pass-through affects borrowers' debt service burden (DSB) and default probability.
- **Key Insight:** Investigated the transmission asymmetry where variable rates amplify the contractionary effect of monetary tightening on borrower consumption, creating a "Debt Deflation" spiral.

Selected Policy & Economic Commentary

- **Strategic Positioning with a Long-term Vision: Gaining the Upper Hand in Global Transformations**
Wen Wei Po (文匯報), Apr 2024 Co-authored with Prof. Zhenhua Mao
 - Analyzed the structural shifts in globalization and proposed strategic layouts for Hong Kong to maintain its competitive edge amidst geopolitical uncertainties.
 - **Key Insight:** Emphasized the necessity of long-term structural reforms over short-term stimulus to navigate global supply chain restructuring.
- **An Independent Force amidst Disruption: Hong Kong's Steady Path against Deglobalization**
Hong Kong International Finance Review, Issue No. 025 Co-authored with Prof. Zhenhua Mao
 - Investigated Hong Kong's role as a resilient "super-connector" in the face of deglobalization headwinds and external economic shocks.
 - **Key Insight:** Argued that Hong Kong's unique institutional advantages allow it to serve as a stabilizing "independent force" in the fragmenting global financial system.